

The `aggregate()` function accepts the following arguments:

- **x**: The object to apply the summary on.
- **by**: A list of elements to be categorised.
- **FUN**: The function to use to compute the statistic.
- **nfrequency**: The number of observations per unit time.
- **...**: Any arguments passed to the function.

The `rowsum()` function can sum the variables in a data set.

For example, the sum of the 'day' column grouped by the job category is shown below:

```
> rowsum(x=bank$day, group=bank$job)
      [,1]
admin.    7803
blue-collar 14646
entrepreneur 2563
housemaid  1713
management 15751
retired    3578
self-employed 2961
services   6470
student    1377
technician 12429
unemployed 2060
unknown    602
```

You can count the number of observations for each value using the `tabulate()` function, as follows:

```
> tabulate(bank$previous)
[1] 286 193 113 78 47 25 22 18 10 4 3 5 1 2
1 0 1 1 1
[20] 1 0 1 1 1 1
```

The `table()` function can also provide a count of the observed data based on categories. For example, for the bank data set, we see the classification of people based on their marital status.

```
> table(bank$marital)
divorced married single
   528    2797    1196
```

The `duplicated()` and `unique()` functions show any repetitive and distinct values for the data. A couple of examples are shown below:

```
> duplicated(new_mtcars$cyl)
[1] FALSE TRUE FALSE TRUE FALSE TRUE
```

```
> unique(bank$job)
[1] "unemployed" "services" "management" "blue-collar"
```

```
[5] "self-employed" "technician" "entrepreneur" "admin."
[9] "student" "housemaid" "retired"
"unknown" ##+END_SRC
```

The summary results can be sorted using the `sort()` function.

```
> r <- rowsum(x=bank$day, group=bank$job)
> r
      [,1]
admin.    7803
blue-collar 14646
entrepreneur 2563
housemaid  1713
management 15751
retired    3578
self-employed 2961
services   6470
student    1377
technician 12429
unemployed 2060
unknown    602

> sort(r)
[1] 602 1377 1713 2060 2563 2961 3578 6470 7803
12429 14646 15751
```

You can also specify the ordering required for a specific column in the data using the `order()` function:

```
> data[order(data$hp), ]
      mpg cyl disp hp drat wt qsec vs am
gear carb minsec
Datsun 710 22.8 4 108 93 3.85 2.320 18.61 1 1
4 1 0.3101667
Valiant 18.1 6 225 105 2.76 3.460 20.22 1 0
3 1 0.3370000
Mazda RX4 21.0 6 160 110 3.90 2.620 16.46 0 1
4 4 0.2743333
Mazda RX4 Wag 21.0 6 160 110 3.90 2.875 17.02 0 1
4 4 0.2836667
Hornet 4 Drive 21.4 6 258 110 3.08 3.215 19.44 1 0
3 1 0.3240000
Hornet Sportabout 18.7 8 360 175 3.15 3.440 17.02 0 0
3 2 0.2836667
```

You are encouraged to read the R documentation for the above functions and try it out on your data sets. 

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